Chapter 8
A Chinese Approach to Learning? A Comparative Study on Time Use Patterns of 15-Year-Old Students in PISA 2012

Yisu Zhou and Dan Wang

Abstract Under the influence of Confucianism, the Chinese society puts more emphasis on effort than on innate ability to account for academic success or failure. As time for learning is a strong indicator of effort, it is expected that Chinese students would spend longer time on learning than their peers in other educational systems. The time use pattern may be an important way to distinguish the Chinese education from educational traditions in other cultures. In this study, we analyze the time use patterns of 15-year-old students in 18 high-performing education systems in PISA 2012. The data show commonalities among four Chinese societies—Hong Kong, Taiwan, Macao, and Shanghai—that students in all four Chinese societies spend long hours on deliberate learning in school, with a focus on reading, mathematics, and science. This pattern distinguishes them from many OECD countries. Additionally, our analysis also reveals intracultural variation. Outside the regular school hours, students from Shanghai and Taiwan dedicate a considerable amount of time to out-of-school-time lessons. In particular, Shanghai students spend much more time on unguided homework, whereas in Hong Kong, Macao, and Taiwan, more importance is given to guided homework and tutoring. In sum, our findings suggest that the Chinese tradition of effort-emphasizing learning approach continues in the modern time, sustained both in and out of school. The extra learning outside school, however, exhibits a new trend indicating the decline of parental guidance.

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8.1 Introduction

In 2009, Shanghai students, representing mainland China to participate for the first time in the Programme for International Student Assessment (PISA), amazed the world by their phenomenal achievements in all three tested subjects, outperforming their peers in other countries by exceptionally large margins. With the stereotypical image of rote learning and exam drilling in China's schools, some commentators remained suspicious about the results by questioning the validity of the assessment in measuring true abilities in the tested domains. Three years later, Shanghai responded to the skepticism with another victory in PISA 2012 for its outstanding performance. The two rounds of PISA results have attracted attention from policymakers as well as educational researchers and stimulated another round of keen interest in educational practices in China (Tan 2013; Tucker 2011).

In fact, there has long been research interest in the USA to understand the Chinese model of education in order to explain the learning gaps between China and the USA. In the 1990s, Stevenson and Stigler (1992) pioneered the comparison between the American, Japanese, and Chinese education systems. Schools in Taiwan were found to be more efficient in the sense that children could achieve higher learning outcomes even with larger class sizes, usually 38–60 compared with the sizes of 20 or below in Western schools. Differences in belief and culture arguably accounted for the learning effectiveness. One of the fundamental distinctions lied in the emphasis on ability or on effort. The American society was inclined to attribute learning outcomes to children's innate ability, whereas Chinese and Japanese parents, teachers, and children were found to believe that learning gaps were primarily caused by various levels of children's effort. Stevenson and Stigler (1992) argued that such emphasis on effort derived from the Confucian philosophy, which rejects categorization of human beings by innate good or bad nature and is interested in moral perfection of all individuals. This argument is echoed by later studies that educational systems under the influence of Confucianism share the emphasis on effort more than innate ability (Tweed and Lehman 2002; Biggs 1998). Similar patterns are discovered among the Asian Americans in the USA recently. Hsin and Xie (2014) find that academic effort exerted by Asian-American students best explains the learning advantages of Asian over White Americans and that such differences in academic effort is further attributed to the cultural beliefs of the two groups in the connection between effort and achievement.

The belief in effort may affect school achievement through the function of learning time. Time was identified as a crucial factor influencing student achievement as early as in the 1960s in the Carroll model (Carroll 1963, 1989). In addition to *aptitude*, *quality of instruction,* and *ability to understand instruction*, student learning is subject to the impact of *opportunity to learn* and *perseverance*. Opportunity to learn refers to the amount of time allowed for learning, for example, in the school sched-

ule. Perseverance is the amount of time a student is willing to spend on learning, which can be indicated by the student's learning time after school, such as homework and supplementary tutoring. Indeed, empirical evidence shows that time for learning, a variable combining "time on task," "opportunity to learn," and "homework," has the largest effect on academic achievement (Seidel and Shavelson 2007). In contrast, reforms that erode instructional time in school adversely influence students' learning in urban school district in the USA (Smith 2000). Further, reducing instructional time in school diminishes students' opportunity in mastery of complex concepts (Clark and Linn 2003). Therefore, the Carroll model offers time as a useful parameter to understand the academic advantages of Chinese students. As time invested in learning is a direct manifestation of effort, Chinese students are expected to spend longer time on learning both within and outside school.

The proposition has been confirmed by some empirical evidence. With the Confucian culture promoting effort, Chinese educational systems provide longer learning time by design, in terms of school schedule. In the 1990s, Chinese elementary children were estimated to spend 240 days or between 1400 and 1700 h a year in school, compared with only 180 days or 1100 h of their American counterparts (Stevenson and Stigler 1992, p. 142). The total learning time at school was about one to two school years longer in Chinese elementary schools than in American schools.

Additionally, the learning time extends to after-school period in the form of homework. Chinese students were found to spend approximately 13 h per week on homework compared with about 4 h per week for American students (Stevenson and Stigler 1992, p. 55). It was common that Chinese teachers assigned homework for students to complete even in winter and summer vacations (Chou and Ching 2012), a practice that would be "inconceivable" for American students (Stevenson and Stigler 1992, p. 55).

On top of the compulsory learning time and activities required by school, Chinese students were also found to be active in participating in cram schools or supplementary tutoring. The expansion of the tutoring industry is particularly conspicuous in Chinese societies, such as Hong Kong, Taiwan, and the Mainland China. In the Mainland China, a survey of 4472 urban households has found 73.8 % of primary students receiving tutoring versus 65.5 % and 53.5 % of junior and senior secondary students (Xue and Ding 2009). Similarly, a 2001 survey of 20,000 middle school students in Taiwan, China, indicated 72.9 % of grade 7 children received tutoring for 6.5 h each week on average (Liu 2012). Participation in supplementary tutoring indicates the commitment to learning and the effort that students are willing to make in academic achievement (Bray 1999; Bray and Lykins 2012).

In sum, time for learning may be a distinctive feature of the Chinese educational system because of the strong Confucian belief in effort. Though student time usage on learning activities has been extensively studied cross nationally in western societies (Dettmers et al. 2009; Trautwein 2007; Trautwein and Köller 2003; Trautwein and Lüdtke 2007), insufficient research attention was devoted understanding

whether time plays a unique role in the Chinese way of education. The citations above show that the specific statistics are available only for the 1990s and only Taiwan was selected as the representative of Chinese education in Stevenson and Stigler's study. Has the pattern identified by Stevenson and Stigler persisted today and throughout various societies in Greater China? How do Chinese students become more similar or different to their peers in other non-Confucian heritage societies in the new context of globalization and convergence of educational policies? This chapter will explore the answers to these questions through comparisons both within different Chinese societies and between Chinese and other Asian and Western countries.

8.2 Data

We used data from the Programme for International Student Assessment 2012 (PISA 2012). PISA is one of the largest international comparative projects in education, conducted by the Organization for Economic Cooperation and Development (OECD). It tests the skills of 15-year-old students in three subjects: mathematics, reading, and science. Four Chinese societies participated in PISA 2012: Hong Kong, Macao, Taiwan, and Shanghai. PISA utilized a two-stage stratified sampling design to collect representative data on 15-year-olds who are still enrolled in schools (OECD 2014). In the first stage, schools were selected systematically with probability proportional to size. In the second stage, at least 35 students were sampled with equal probability within each selected school (OECD 2014).

8.2.1 Countries for Comparison

In addition to the four Chinese societies, we also include several other countries as points of comparison in our analysis. The criterion of selection is high performance in the PISA 2012 assessment. We select countries that scored consistently above the OECD average in all three subject areas. By forming our baseline from a comparison of high-performing countries, we can develop a better understanding of the role of time use in shaping Chinese students' learning process. Our selection criterion leaves us with four Asian countries and ten OECD countries (Table 8.A1).

8.2.2 Variables

In our study, we focus on two sets of time usage variables. The first set of variables measures time usage on what the OECD calls "deliberate learning activities" (OECD 2011, p. 20), particularly regular classes on the tested academic subjects of

reading, math, science, as well as on other subjects. The second set of variables measures time usage on additional learning activities that take place outside classroom settings, including out-of-school-time lessons and what we call "after-school learning activities." While time use on deliberate learning activities is driven by structural forces such as the length of the school day and is thus seen by some as a more suitable tool for addressing educational equality issues (OECD 2011), learning activities outside classrooms, our second set of measures, are equally important and, in our view, heavily influenced by the sociocultural milieu. Competitive exit exams may contribute to the demand of tutoring lessons for many students, while class-specific preference may lead to differentiated participation in enrichment or remedial type of activities. The role these activities play in shaping learning outcomes in Chinese societies is not clearly understood.

Both sets of variables are taken from the PISA student questionnaire. The variables in the first set are `st69` and `st70`. These items ask students to recall the length in minutes of a typical class period and the number of class periods they have each week. We use such information to calculate the number of weekly instructional hours on different subjects.

The second set of variables consists of `st55` and `st57`. There are four subquestions for `st55`, which ask students to estimate the number of hours they spent on out-of-school lessons per week. The original items are categorical; we have converted them to a continuous format using the method described in another study (Zhou and Wang 2015). The six subquestions for `st57` ask students to estimate the number of hours usually spent each week on additional learning tasks. We sort these activities into four broad categories: unguided homework, guided homework, studying with parents, and receiving tutoring. These two items measure overlapping, albeit different, types of activities: the out-of-school lesson questions focus on lessons as the major form of activity, regardless of the nature of such lessons, while the after-school learning activity questions cover a wider range of activities and tasks with a specific purpose.

The definition of each variable is given in Appendix Table 8.A2.

8.3 Analysis

8.3.1 Learning Time in School

We first examine student time use inside schools (i.e., deliberate learning activities). We distinguish two types of time use: that spent on the tested subjects (reading, mathematics, and science) and that spent on other classes. This distinction yields a typology of countries with four categories (see Fig. 8.1):

1. More absolute learning time overall and more relative learning time on tested subjects

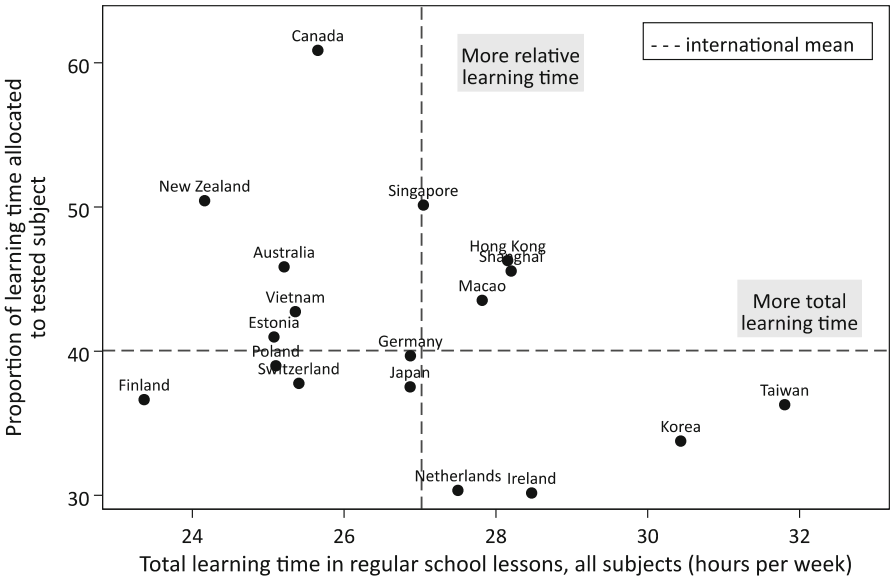

Fig. 8.1 Relative learning time on testing subjects versus absolute learning time in regular class hours

2. More absolute learning time overall and less relative learning time on tested subjects
3. Less absolute learning time overall and more relative learning time on tested subjects
4. Less absolute learning time overall and less relative learning time on tested subjects

First, all four Chinese societies are characterized by "more absolute learning time," since the amount of regular school hours in all four surpassed the international average of high-performing education systems (i.e., all four lay to the right of the vertical x-reference line). Among the four regions, Taiwan registered the highest number of hours (31.8). The long deliberate learning hours are not unique to Chinese societies; they are also common in other Asian countries such as Korea, Singapore, and Japan.

Second, beyond their longer hours, Chinese education systems tend to have a clear focus on the tested subjects (Category 1), except for Taiwan (Category 2). It should be noted that the PISA definition of the tested subjects differs from actual tested subjects in most Chinese societies, where English is most likely considered a core academic subject alongside reading, math, and science. Therefore, by using PISA's definition, we actually underestimate the proportion of deliberate learning time devoted to academic subjects. One would expect an even stronger focus on

academic learning in Chinese societies in practice. The pattern of time allocation within deliberate learning activities in Singapore is very similar to that in Hong Kong, Shanghai, and Macao. In contrast, the pattern in Taiwan has more resemblance with that of Korea than with the patterns in the other Chinese societies.

This feature of Chinese education systems contrasts with high-performing OECD education systems, many of which have shorter deliberate learning hours and some of which even deemphasize tested subjects (Finland, Switzerland, and Poland, falling in Category 4). On the other hand, Canada, New Zealand, and Australia clearly follow the "shorter class hours but more focus on academics" model (Category 3). Students in these countries spend a higher proportion of their class time on academic subjects.

8.3.2 Learning Time Outside School Schedule

For the next aspect we analyze the time invested in after-school learning activities. We find that students in Chinese societies are likely to put in considerable extra effort outside of regular school time. This is illustrated in Fig. 8.2. Focusing on the three tested subjects, we compare the amount of time students in each system spend on these subjects in school and out of school. Again, we can categorize the systems into four types:

1. More learning time on tested subjects in school and out of school
2. More learning time on tested subjects in schools but less out of school

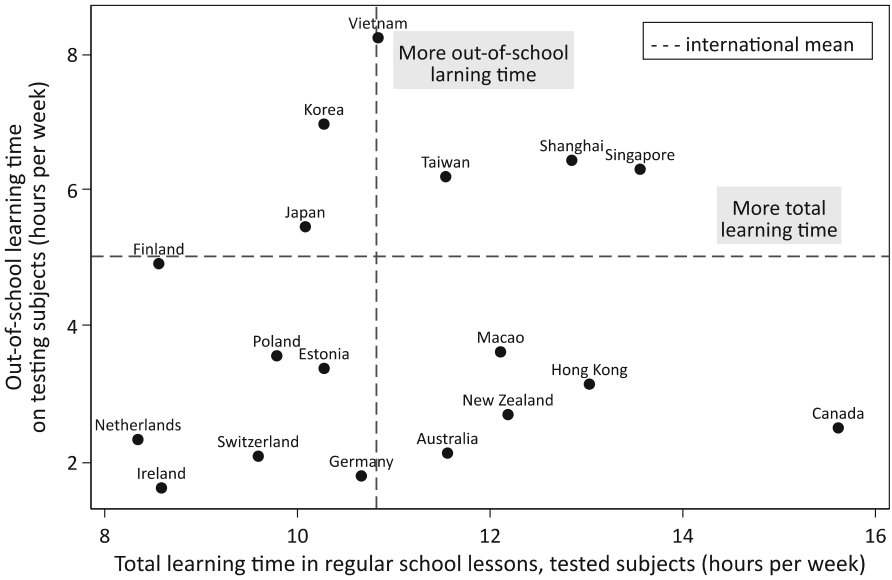

Fig. 8.2 Time spent on tested subjects outside school and in regular school time

3. Less learning time on tested subjects in schools but more out of school
4. Less learning time on tested subjects in schools and out of school

Consistent with previous findings, students in all four Chinese societies spend a relatively large amount of deliberate learning time on the tested subjects. A difference is apparent in the time spent on extra lessons, with students in Taiwan and Shanghai taking more extra lessons in the tested subjects than students in Macao and Hong Kong. But even in the latter two regions, the time spent on extra lessons surpassed that in the majority of OECD countries. The time spent on extra lessons by students in Singapore is comparable to the time spent by students in Taiwan and Shanghai, and it should not come as a surprise that all the countries with above-average extra lesson consumption are Asian.

Based on Figs. 8.1 and 8.2, we conclude that the four Chinese societies share the common feature of long hours in regular school lessons. However, there are also important differences. Shanghai not only has a more clear focus on tested subjects in schools, but its students also put in considerable extra effort out of school. Hong Kong and Macao share the strong focus on tested subjects in schools, but their students take fewer out-of-school lessons in these subjects. Taiwan displays the opposite pattern: while schools deemphasize tested subjects, students are compensating through lessons outside schools.

The limitation of the out-of-school lesson rubric is that it does not reveal the nature of the extra learning efforts. Next we analyze how students spend their time in the course of after-school learning activities (see Table 8.1). These learning activities most likely occur after the end of the formal school day, and they can take a variety of forms. We break down the average number of hours spent weekly on each type of activity and the proportion that this represents of total time usage on after-school learning activities.

In this analysis, Shanghai stands out as unique, with Shanghai students spending the most time on after-school learning, 18.15 h per week. This high figure is due to long hours spent on unguided homework. On average, Shanghai students spend 11.66 h per week, more than 2 h per day, working on homework by themselves. However, in terms of relative time allocation Shanghai is not unique: the time spent on unguided homework accounts for 64 % of total after-school efforts in Shanghai, on par with the proportion in several other systems such as Finland, Germany, Ireland, and the Netherlands. It is the absolute time expenditure that distinguishes Shanghai students from those of other high-performing systems.

In addition, students in Shanghai seek guidance to complement their solitary work by spending an extra 2.5 h per week doing guided homework. This finding again distinguishes Shanghai from other education systems, even its Asian or Chinese neighbors. For instance, students in Vietnam also spend long hours after school on learning, but they are primarily investing in tutoring activities (personal or with a commercial company, Table 8.A2). Rarely do students in Vietnam work on their homework unguided. Singaporean students mirror Shanghai students in terms

Table 8.1 Time usage on after-school learning activities (absolute hours and relative allocation)

Designation	Country/region	Unguided homework	Guided homework	Studying with parents	Receiving tutoring	Total
Chinese societies	Hong Kong	4.98	1.23	0.50	1.70	8.41
		59 %	15 %	6 %	20 %	100 %
	Macao	4.30	1.96	0.56	1.67	8.49
		51 %	23 %	7 %	20 %	100 %
	Shanghai	11.66	2.49	0.76	3.23	18.15
		64 %	14 %	4 %	18 %	100 %
	Taiwan	4.34	1.27	0.90	2.18	8.68
		50 %	15 %	10 %	25 %	100 %
Asian countries	Japan	3.15	0.78	0.28	0.67	4.87
		65 %	16 %	6 %	14 %	100 %
	Korea	2.11	0.85	0.41	4.96	8.34
		25 %	10 %	5 %	60 %	100 %
	Singapore	7.57	2.41	0.87	3.03	13.89
		55 %	17 %	6 %	22 %	100 %
	Vietnam	3.78	2.89	1.75	6.46	14.88
		25 %	19 %	12 %	43 %	100 %
OECD countries	Australia	4.98	1.27	1.03	0.78	8.06
		62 %	16 %	13 %	10 %	100 %
	Canada	4.45	1.24	0.87	0.72	7.27
		61 %	17 %	12 %	10 %	100 %
	Estonia	5.45	1.54	0.92	1.42	9.32
		58 %	17 %	10 %	15 %	100 %
	Finland	2.31	0.47	0.44	0.19	3.41
		68 %	14 %	13 %	5 %	100 %
	Germany	4.48	0.20	1.02	1.05	6.75
		66 %	3 %	15 %	16 %	100 %
	Ireland	5.69	1.64	0.67	0.63	8.63
		66 %	19 %	8 %	7 %	100 %
	Netherlands	5.00	0.99	1.01	0.73	7.73
		65 %	13 %	13 %	9 %	100 %
	New Zealand	3.37	0.96	0.82	0.63	5.78
		58 %	17 %	14 %	11 %	100 %
	Poland	4.65	1.95	1.24	1.75	9.59
		48 %	20 %	13 %	18 %	100 %
	Switzerland	3.15	0.92	0.98	0.63	5.67
		55 %	16 %	17 %	11 %	100 %

of time allocation on additional learning activities. The distribution of time use on such activities is similar in the two systems, but students in Shanghai outspend their Singaporean counterparts in the unguided homework category by 4 h per week.

Students in Hong Kong, Macao, and Taiwan do not differ much from students in other high-performing systems, either in absolute terms or in terms of relative time allocation. The only variation is in the time spent on tutoring. The average number of hours spent on tutoring is generally higher in the three Chinese societies than it is in OECD countries, but lower than in most of their Asian neighbors.

Examining students' time usage on various after-school learning activities is important to understand how students orchestrate their life outside school. In Fig. 8.3, we look at the amount of time students in each region spend on each activity category in relation to regular class time (expressed as a percentage). We stress the linkage between deliberate learning time and after-school learning time because additional learning is always relative. In many systems, after-school learning is a direct reflection of what students are taught in classrooms. Students may seek additional clarification of what they have been taught at school, or they may be compelled to review school lessons and to complete their homework. The numbers in Fig. 8.3 are a reflection of extra effort made outside of regular academic classes. Shanghai, Singapore, and Vietnam are the three education systems in which students spend at least 50 % more time on learning beyond regular class hours. But there are two important distinctions.

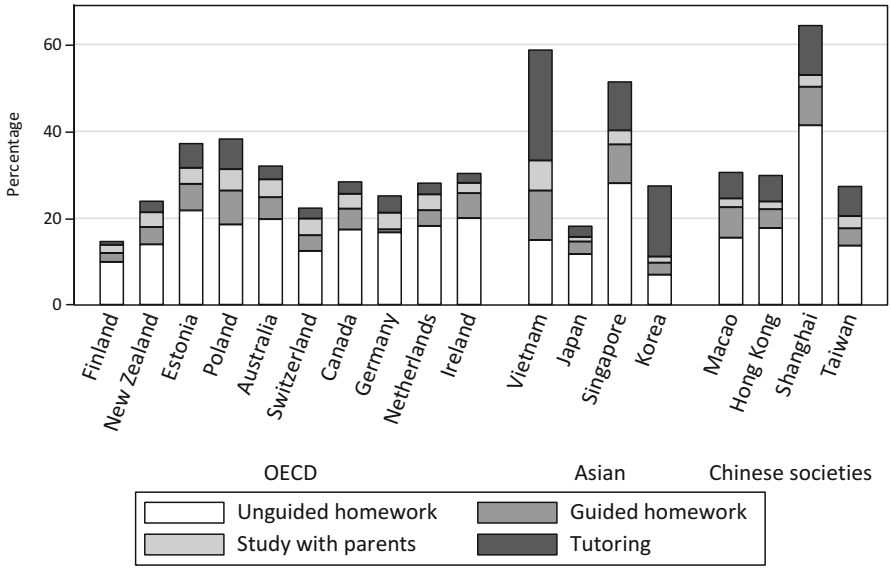

Fig. 8.3 Relative allocation of time on after-school learning activities in relation to regular school hours

First, students' total in-school learning time in Vietnam is lower than that in Singapore and Shanghai (Fig. 8.1). This suggests that the Vietnamese students are likely to use after-school programs to supplement what they have learned in school. In Singapore and Shanghai, by contrast, the after-school learning activities come on top of already long learning hours in school. In the Chinese context, this may reflect a pattern of putting the icing on the cake. Second, the patterns of time allocation are different in the three societies. In Shanghai, the bulk of the additional learning involves doing teacher-assigned homework. This scenario applies also to Singapore, though to a lesser degree. But the additional learning of students in Vietnam, probably reflecting its supplementary nature, is centered on tutoring activities, either with an individual or with a commercial company.

One commonality among Chinese societies is the reduced role of the parents compared to OECD countries, as indicated by the smaller proportion of time that Chinese students allocate to studying with their parents. It is possible that the greater popularity of guided learning centers and other complementary services outside schools has left parents feeling less compelled to take up the traditional role of their child's learning partner.

8.4 Discussion

In this chapter we have investigated students' use of learning time in four Chinese societies and compared it with patterns in other high-performing Asian and OECD countries. Our analysis reveals some unique characteristics of time use on learning activities in Chinese systems, but it also suggests that there is equally important intracultural variation. In some ways, the four Chinese societies are similar. The absolute amount of time spent on deliberate learning is high in all of these societies. Three out of the four systems have a specific focus on PISA-tested subjects, and the number of hours spent on tested subjects in the fourth system, Taiwan, also exceeds the international average. The combination of these two features distinguishes Chinese societies from other high-performing nations.

In addition to spending long hours on deliberate learning in school, the data show that students in two Chinese societies (Taiwan and Shanghai) also put in extra effort in out-of-school-time lessons. Scholars attribute the academic success of the overseas Chinese students from these societies to such activities (Hsin and Xie 2014). If we examine the relative allocation of time among different types of after-school learning activities, differences among Chinese societies emerge.

Shanghai students invest their time mostly in unguided homework, while students in Hong Kong, Macao, and Taiwan spend more time on guided homework and tutoring. Overall, tutoring has become increasingly important in Chinese societies, as students seek a second source of authoritative knowledge. Chinese societies are similar to other Asian societies in this regard, as the proportion of after-school

learning time devoted to tutoring is comparable in both groups and generally higher than in OECD countries. Unfortunately, the PISA data do not elucidate the nature of these tutoring classes. Students also seem to be seeking guidance on their homework, but they are doing so outside the traditional parent-student relationship. They may be working with each other, with senior students, or even with teachers. The prevalence of alternative avenues for learning could explain why studying with parents is less important for Chinese students than for students in OECD countries.

Time use pattern is linked with two aspects of student learning. On one hand, deliberate learning time is a direct reflection of institutional design of formal schooling inside traditional classroom, while on the other hand, after-school learning time represents personal investment in education taking alternative forms. The Chinese model of education thus reflects not only societal-wide emphasis but also echoes individual pursuit of education excellence. Yet from a time use perspective, this model of education, though it takes certain unique manifestation in Chinese societies, is found in other Asian systems.

Our analysis also leads us to single out the case of Shanghai. Shanghai students, ranked as the highest performers in PISA 2012, spend the most time on extra learning. In addition to classroom learning, Shanghai students put in an additional 18 h of learning per week, equivalent to extending school hours by 60 %. This learning pattern distinguishes Shanghai from all other education systems in our sample. On average, 15-year-old Shanghai students spend more than 14 h every week on homework. That is more than 2 h on a typical school day. According to the PISA questionnaire, this homework is by definition set by teachers. This means that the influence of school extends beyond regular class hours. The schooling experience in Shanghai (and to a lesser degree in other Chinese societies) must thus be seen as a continuous process that can take place in multiple places. This is a unique pattern and one that we believe is worth further investigation in the future. Though the emergence of a plethora of learning services could prompt the shift of locus of learning from schools to other venues, the story we have uncovered seems to suggest that school-initiated learning activities still dominate in Chinese societies. In a sense, then, the trend in these societies represents a reinforcement of the old model, not the invention of a novel one.

8.5 Limitations of This Study

Though PISA provides a nationally/regionally representative sample of 15-year-olds, we caution readers of potential perils caused by the quality of the data. Our concern stems largely from the instrument. The time use information is self-reported by students. Errors may occur when students attempt to recall their past time expenditure on various activities. Additionally, as Bray and Kobakhidze (2014) have pointed out, seasonal variations in participation patterns in certain activities are not captured by the PISA questionnaire. Therefore, one should expect some measurement errors in the statistics we report.

Other issues arise from the cross-cultural translation of the survey items. We considered this problem in an earlier study (Zhou and Wang 2015), and it is worth further discussion here. While in English terms such as "homework" and "studying" may refer fairly unambiguously to distinctive learning activities, there is little guarantee that such distinctions are preserved in cross-cultural translation. To our knowledge, the translation of survey items from English to different forms of Chinese is conducted by four independent PISA centers, and therefore the language used in each version may be different. Such divergence was indicated by our comparison of item wording across pilot versions of the student questionnaire. The comparison revealed differences in words used, potentially conveying different meanings. However, our request to examine final versions of the student questionnaire was not granted, and we were thus not able to confirm whether such differences were resolved in the survey. Since our analysis is secondary and we are not able to validate the data ourselves, we have taken the questionnaire items at face value and categorized them in ways that we believe are appropriate and meaningful. We also place our faith in the OECD quality assurance procedure. It is nonetheless possible that our categories are not mutually exclusive and that one may encompass part of another.

Appendix

Table 8.A1 Countries/regions of comparison in the sample

Designation	Country/region	PISA performance		
		Reading	Math	Science
Chinese societies	Hong Kong	545	561	555
	Macao	509	538	521
	Shanghai	570	613	580
	Taiwan	523	560	523
Asian countries	Japan	538	536	547
	Korea	536	554	538
	Singapore	542	573	551
	Vietnam	508	511	528
OECD countries	Australia	512	504	521
	Canada	523	518	525
	Estonia	516	521	541
	Finland	524	519	545
	Germany	508	514	524
	Ireland	523	501	522
	Netherlands	511	523	522
	New Zealand	512	500	516
	Poland	518	518	526
	Switzerland	509	531	515

Table 8.A2 Definitions of time use variables

Name	Definition	Construction
Total class periods	Total hours spent per week in class (minutes per class × number of class periods)	st69q01*st71/60
Tested subjects	Total hours spent on reading, math, and science classes	(st69q01*st70q01 + st69q02*st70q02 + st69q03*st70q03)/60
Non-tested subjects	Total hours spent on classes other than reading, math, and science	(st69q01*st71−st69q01*st70q01 −st69q02*st70q02st69q03*st70q03)/60
Out-of-school time lessons	Total hours spent on out-of-school-time lessons[a]	st55q01 + st55q02 + st55q03 + st55q04
Unguided homework	Homework or other study set by teachers and not carried out with somebody overlooking and providing help, either at school or elsewhere	st57q01−st57q02
Guided homework	Homework or other study set by teachers and done with somebody overlooking and providing help, either at school or elsewhere	st57q02
Study with parents	Study with a parent or another family member	st57q05
Tutoring	Work with a personal tutor (whether paid or not) or attendance in out-of-school classes provided by a commercial company and paid for by parents	st57q03 + st57q04

[a]After the conversion of categorical variable to a continuous scale

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